Goal relaxation approach

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Goal: Anticipate the impact of floods on people

I. Calculate similarity of each triple with the expressed goal. Rank triples based on best score.

TRIPLES	GOAL	SCORE
Flood Causes Health and safety risks	Anticipate the impact of floods on people	0.71685076
Flood Causes Loss of life	Anticipate the impact of floods on people	0.6742301
Flood Causes Environmental damage	Anticipate the impact of floods on people	0.64711046
Flood warning system Analyzes Forecasting	Anticipate the impact of floods on people	0.64173543
Flood Causes Business disruption	Anticipate the impact of floods on people	0.6296501
Flood Causes Property damage	Anticipate the impact of floods on people	0.61910564
Flood Causes Evacuation of residents	Anticipate the impact of floods on people	0.6152678
Flood warning system Predicts Flood	Anticipate the impact of floods on people	0.610021
Flood warning system Recommends Evacuation of residents	Anticipate the impact of floods on people	0.6083703
Sensors Detect Flood	Anticipate the impact of floods on people	0.56785285
Heavy rainfall Causes Flood	Anticipate the impact of floods on people	0.5578227
Flood Causes Traffic disruption	Anticipate the impact of floods on people	0.5335632
Dam failure Causes Flood	Anticipate the impact of floods on people	0.53266644
Cameras Detect Flood	Anticipate the impact of floods on people	0.5231204
Dam Provides Flood control	Anticipate the impact of floods on people	0.52188134
Tsunami Causes Flood	Anticipate the impact of floods on people	0.51081204
Flood warning system Generates Alerts	Anticipate the impact of floods on people	0.49050453
River overflow Causes Flood	Anticipate the impact of floods on people	0.47856724
Dam Avoids Flood	Anticipate the impact of floods on people	0.46423432
Weather prediction Anticipates Heavy rainfall	Anticipate the impact of floods on people	0.45595518
Sensors Measure Heavy rainfall	Anticipate the impact of floods on people	0.40247208
Dam failure Due to Inadequate maintenance	Anticipate the impact of floods on people	0.39204824
Emergencies response agencies is a type of General public	Anticipate the impact of floods on people	0.38158426
Dam failure Due to Overtopping	Anticipate the impact of floods on people	0.3579417
Dam Fails Dam failure	Anticipate the impact of floods on people	0.34935087

Earthquake Causes Tsunami	Anticipate the impact of floods on people	0.33516264
Dam Provides Irrigation	Anticipate the impact of floods on people	0.32057822
Homes damage is a type of Property damage	Anticipate the impact of floods on people	0.31964794
Dam failure Due to Material defects	Anticipate the impact of floods on people	0.3182178
Buildings damage is a type of Property damage	Anticipate the impact of floods on people	0.30909312
Siphoning Causes Flood	Anticipate the impact of floods on people	0.29804587
Weather forecasting system Analyzes Data	Anticipate the impact of floods on people	0.2933868
Weather forecasting system is a type of Weather prediction	Anticipate the impact of floods on people	0.2789752
Railroads damage is a type of Property damage	Anticipate the impact of floods on people	0.2666812
Bridge damage is a type of Property damage	Anticipate the impact of floods on people	0.26455322
Sensors Measure Water level	Anticipate the impact of floods on people	0.2610495
People is a type of General public	Anticipate the impact of floods on people	0.25760114
Weather forecasting system Generates Forecasting	Anticipate the impact of floods on people	0.24966733
Emergency radio broadcasts is a type of Communication channels	Anticipate the impact of floods on people	0.24857144
Sensors Measure Streamflow	Anticipate the impact of floods on people	0.22437562
Vehicles damage is a type of Property damage	Anticipate the impact of floods on people	0.21851136
Sensors Measure Soil moisture	Anticipate the impact of floods on people	0.2094015
Government authorities is a type of General public	Anticipate the impact of floods on people	0.18049017
Alerts Issued to Communication channels	Anticipate the impact of floods on people	0.1738446
Alerts Notify General public	Anticipate the impact of floods on people	0.1692378
Sirens is a type of Communication channels	Anticipate the impact of floods on people	0.14984675
Mobile phone messages is a type of Communication channels	Anticipate the impact of floods on people	0.13214621
Social media notification is a type of Communication channels	Anticipate the impact of floods on people	0.12852414
Television announcements is a type of Communication channels	Anticipate the impact of floods on people	0.1194395
Sensors Generate Data	Anticipate the impact of floods on people	0.10778736
Cameras Generate Data	Anticipate the impact of floods on people	0.08312291
Dedicated smartphone app is a type of Communication channels	Anticipate the impact of floods on people	0.07879389
Consumer cameras is a type of Cameras	Anticipate the impact of floods on people	0.0300335
Remote cameras is a type of Cameras	Anticipate the impact of floods on people	0.012958915

II. Identify anchor triples ranked above a threshold.

Select the best triples as anchor points by applying a threshold **[0.85*best score, best score]**.

TRIPLES	GOAL	SCORE
Flood Causes Health and safety risks	Anticipate the impact of floods on people	0.71685076
Flood Causes Loss of life	Anticipate the impact of floods on people	0.6742301
Flood Causes Environmental damage	Anticipate the impact of floods on people	0.64711046
Flood warning system Analyzes Forecasting	Anticipate the impact of floods on people	0.64173543
Flood Causes Business disruption	Anticipate the impact of floods on people	0.6296501
Flood Causes Property damage	Anticipate the impact of floods on people	0.61910564
Flood Causes Evacuation of residents	Anticipate the impact of floods on people	0.6152678
Flood warning system Predicts Flood	Anticipate the impact of floods on people	0.610021

III. Carry out sentiment analysis and add 'Prevent that' to negatively qualified anchor triples.

The triples with a negative meaning are transformed by using sentiment analysis.
The words "**Prevent that**" are added at the beginning of the negative triples.

TRIPLES (ANCHOR POINTS)	GOAL
Prevent that Flood Causes Health and safety risks	Anticipate the impact of floods on people
Prevent that Flood Causes Loss of life	Anticipate the impact of floods on people
Prevent that Flood Causes Environmental damage	Anticipate the impact of floods on people
Flood warning system Analyzes Forecasting	Anticipate the impact of floods on people
Prevent that Flood Causes Business disruption	Anticipate the impact of floods on people
Prevent that Flood Causes Property damage	Anticipate the impact of floods on people
Prevent that Flood Causes Evacuation of residents	Anticipate the impact of floods on people
Flood warning system Predicts Flood	Anticipate the impact of floods on people
Prevent that Flood Causes Health and safety risks	Anticipate the impact of floods on people

IV. Test entailment between each anchor triples and the goal. Keep entailing anchor triples.

Test the entailment of each **anchor point** with the goal (note: E: Entailment, N: Neutral and C: Contradiction).

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Evacuation of residents	Anticipate the impact of floods on people	**0.86817**	0.11147	0.02036	ENTAILMENT
Prevent that Flood Causes Loss of life	Anticipate the impact of floods on people	**0.78269**	0.17343	0.04388	ENTAILMENT
Prevent that Flood Causes Health and safety risks	Anticipate the impact of floods on people	**0.70614**	0.21396	0.0799	ENTAILMENT
Prevent that Flood Causes Business disruption	Anticipate the impact of floods on people	**0.41095**	0.35846	0.23059	ENTAILMENT
Flood warning system Predicts Flood	Anticipate the impact of floods on people	0.3862	0.59991	0.01389	NEUTRAL
Prevent that Flood Causes Property damage	Anticipate the impact of floods on people	**0.3506**	0.34778	0.30162	ENTAILMENT
Flood warning system Analyzes Forecasting	Anticipate the impact of floods on people	0.21787	0.76808	0.01405	NEUTRAL
Prevent that Flood Causes Environmental damage	Anticipate the impact of floods on people	0.16007	0.38834	0.45158	CONTRADICTION

Keep the entailed triples (anchor points):

- **Prevent that Flood Causes Evacuation of residents**
- **Prevent that Flood Causes Loss of life**
- **Prevent that Flood Causes Health and safety risks**
- **Prevent that Flood Causes Business disruption**
- **Prevent that Flood Causes Property damage**

1) For non-entailing anchor triples, relax the goal by making it more abstract
Keep entailing anchor triples.

Non-entailed triples:

- Flood warning system Predicts Flood
- Flood warning system Analyzes Forecasting
- Prevent that Flood Causes Environmental damage

Note: the tool informs the designer that they may relax the goal to obtain more results.

Relax the goal:
Anticipate the impact of flood on people → Anticipate the impact of flood

Test the entailment:

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Flood warning system Predicts Flood	Anticipate the impact of floods	**0.74647**	0.23932	0.01419	ENTAILMENT

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Flood warning system Analyzes Forecasting	Anticipate the impact of floods	**0.57027**	0.40243	0.02729	ENTAILMENT

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Environmental damage	Anticipate the impact of floods	**0.58834**	0.17329	0.23836	ENTAILMENT

V. Present entailed entailing triples to the designer.

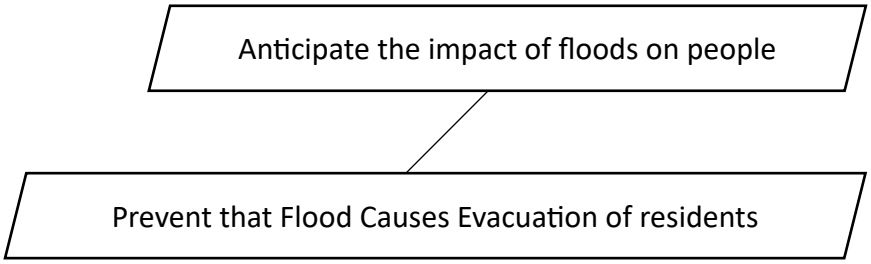
Present each anchor triples together with the corresponding goal (initial or relaxed).

Entailing anchor triples:

Anchor points	Goal (initial and relaxed)
Prevent that Flood Causes Evacuation of residents	Anticipate the impact of floods on people
Prevent that Flood Causes Loss of life	Anticipate the impact of floods on people
Prevent that Flood Causes Health and safety risks	Anticipate the impact of floods on people
Prevent that Flood Causes Business disruption	Anticipate the impact of floods on people
Prevent that Flood Causes Property damage	Anticipate the impact of floods on people
Flood warning system Predicts Flood	Anticipate the impact of floods
Flood warning system Analyzes Forecasting	Anticipate the impact of floods
Prevent that Flood Causes Environmental damage	Anticipate the impact of floods

1) The designer expresses their wish to explore more one or more entailed anchor triples. Designate each of these anchor triples as the new goal and its neighbour triples as the new anchor triples. Start to refine the anchor triple with the highest entailment score. Repeat the approach starting from (III).

a) Prevent that Flood Causes Evacuation of residents



The designer wants to explore more the anchor triple:
Prevent that Flood Causes Evacuation of residents (with **0.86817%**)

Extract the triple neighbors and apply the sentiment analysis on triples:

TRIPLE NEIGHBORS	GOAL
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Evacuation of residents
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Evacuation of residents
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Evacuation of residents
Prevent that Flood Causes Property damage	Prevent that Flood Causes Evacuation of residents
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Evacuation of residents
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Evacuation of residents
Prevent that River overflow Causes Flood	Prevent that Flood Causes Evacuation of residents
Sensors Detect Flood	Prevent that Flood Causes Evacuation of residents
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Evacuation of residents
Dam Avoids Flood	Prevent that Flood Causes Evacuation of residents
Flood warning system Predicts Flood	Prevent that Flood Causes Evacuation of residents
Cameras Detect Flood	Prevent that Flood Causes Evacuation of residents
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Evacuation of residents
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Evacuation of residents
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Evacuation of residents
Flood warning system Recommends Evacuation of residents	Prevent that Flood Causes Evacuation of residents

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Evacuation of residents	0.14775	0.38578	0.46647	CONTRADICTION
Flood warning system Recommends Evacuation of residents	Prevent that Flood Causes Evacuation of residents	0.05466	0.52064	0.4247	NEUTRAL
Flood warning system Predicts Flood	Prevent that Flood Causes Evacuation of residents	0.01835	0.97364	0.00802	NEUTRAL
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Evacuation of residents	0.00955	0.97705	0.01339	NEUTRAL
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Evacuation of residents	0.00712	0.93543	0.05745	NEUTRAL
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Evacuation of residents	0.00594	0.25485	0.73921	CONTRADICTION
Cameras Detect Flood	Prevent that Flood Causes Evacuation of residents	0.00513	0.98685	0.00803	NEUTRAL
Sensors Detect Flood	Prevent that Flood Causes Evacuation of residents	0.00441	0.98197	0.01361	NEUTRAL
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Evacuation of residents	0.0044	0.69172	0.30388	NEUTRAL
Prevent that Flood Causes Property damage	Prevent that Flood Causes Evacuation of residents	0.00422	0.32498	0.6708	CONTRADICTION
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Evacuation of residents	0.0038	0.34612	0.65008	CONTRADICTION
Dam Avoids Flood	Prevent that Flood Causes Evacuation of residents	0.00379	0.99355	0.00267	NEUTRAL
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Evacuation of residents	0.00365	0.16509	0.83126	CONTRADICTION
Prevent that River overflow Causes Flood	Prevent that Flood Causes Evacuation of residents	0.00357	0.98607	0.01036	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Evacuation of residents	0.00296	0.9893	0.00774	NEUTRAL
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Evacuation of residents	0.00252	0.98653	0.01095	NEUTRAL

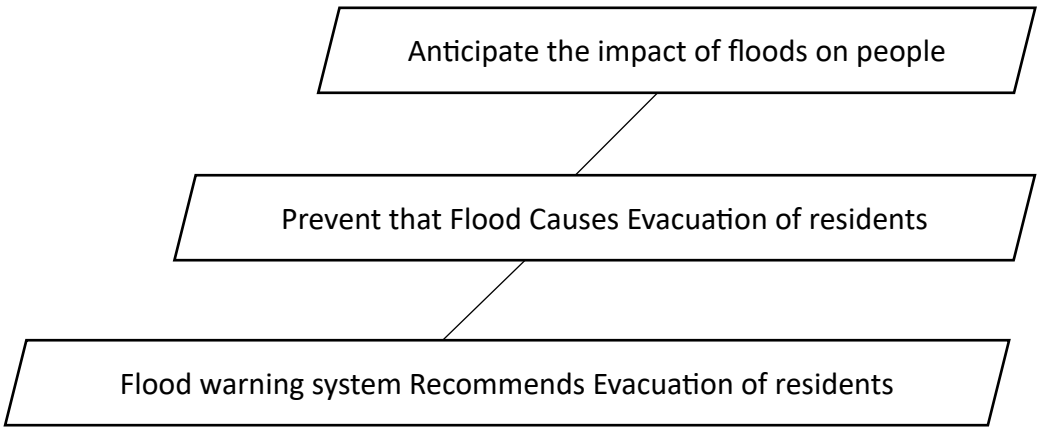
No entailment.

Relax the goal:
Prevent that Flood Causes Evacuation of residents → **Urge the evacuation of residents**
(keep the object in the goal)

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Flood warning system Recommends Evacuation of residents	Urge the evacuation of residents	**0.93647**	0.0457	0.01782	ENTAILMENT
Prevent that Flood Causes Traffic disruption	Urge the evacuation of residents	0.00908	0.44571	0.5452	CONTRADICTION
Prevent that Flood Causes Business disruption	Urge the evacuation of residents	0.00594	0.58774	0.40632	NEUTRAL
Flood warning system Predicts Flood	Urge the evacuation of residents	0.00548	0.26557	0.72895	CONTRADICTION
Prevent that Tsunami Causes Flood	Urge the evacuation of residents	0.00504	0.80681	0.18815	NEUTRAL
Prevent that Flood Causes Property damage	Urge the evacuation of residents	0.00484	0.92328	0.07188	NEUTRAL
Prevent that Flood Causes Loss of life	Urge the evacuation of residents	0.00476	0.94589	0.04936	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Urge the evacuation of residents	0.00474	0.93843	0.05682	NEUTRAL
Prevent that Dam failure Causes Flood	Urge the evacuation of residents	0.00451	0.84794	0.14755	NEUTRAL
Prevent that Flood Causes Health and safety risks	Urge the evacuation of residents	0.00449	0.66845	0.32706	NEUTRAL
Prevent that Siphoning Causes Flood	Urge the evacuation of residents	0.00429	0.86177	0.13394	NEUTRAL
Prevent that Flood Causes Environmental damage	Urge the evacuation of residents	0.00414	0.73982	0.25604	NEUTRAL
Prevent that River overflow Causes Flood	Urge the evacuation of residents	0.00388	0.9168	0.07931	NEUTRAL
Dam Avoids Flood	Urge the evacuation of residents	0.00294	0.38599	0.61107	CONTRADICTION
Cameras Detect Flood	Urge the evacuation of residents	0.00263	0.9034	0.09397	NEUTRAL
Sensors Detect Flood	Urge the evacuation of residents	0.00232	0.95329	0.04439	NEUTRAL

Refine the goal “Prevent that Flood Causes Evacuation of residents” by asking HOW question: How to prevent the evacuation of residents?



Explore the triple neighbors of “Flood warning system Recommends Evacuation of residents” and apply the sentiment analysis. Remove triples that are already entailed (in this branch).

TRIPLE NEIGHBORS	GOAL
Flood warning system Predicts Flood	Flood warning system Recommends Evacuation of residents
Flood warning system Analyzes Forecasting	Flood warning system Recommends Evacuation of residents
Flood warning system Generates Alerts	Flood warning system Recommends Evacuation of residents
~~Prevent that Flood Causes Evacuation of residents~~	~~Flood warning system Recommends Evacuation of residents~~

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Prevent that Flood Causes Evacuation of residents~~	~~Flood warning system Recommends Evacuation of residents~~	~~0.01412~~	~~0.7183~~	~~0.26759~~	~~NEUTRAL~~
Flood warning system Predicts Flood	Flood warning system Recommends Evacuation of residents	0.00779	0.8939	0.09831	NEUTRAL
Flood warning system Generates Alerts	Flood warning system Recommends Evacuation of residents	0.00335	0.91091	0.08574	NEUTRAL
Flood warning system Analyzes Forecasting	Flood warning system Recommends Evacuation of residents	0.00211	0.89351	0.10438	NEUTRAL

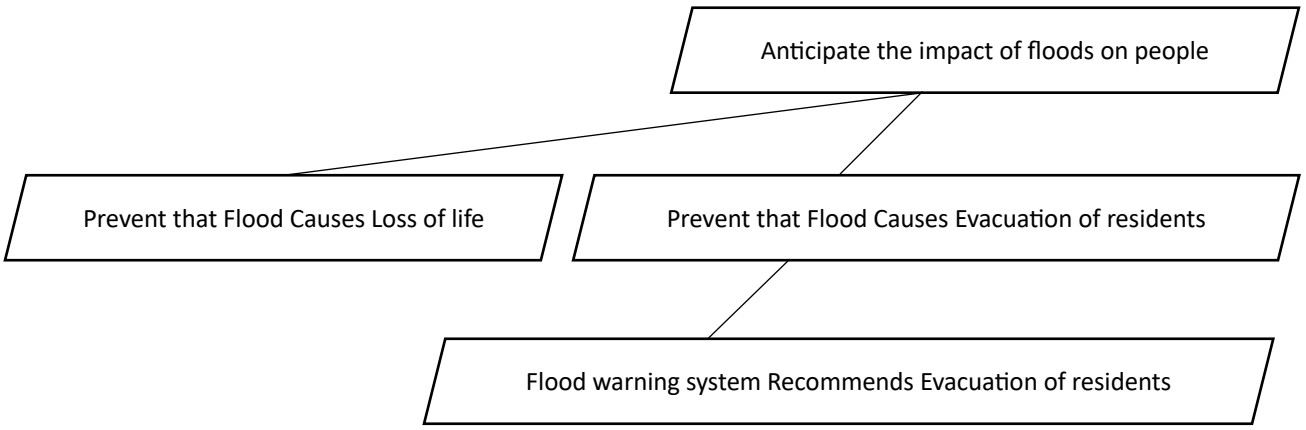
No entailment.

The goal relaxation cannot be achieved. No triples on evacuating residents. It is difficult to make more abstract.

Stop the refinement.

Select the next anchor triple with highest entailment score:

Prevent that Flood Causes Loss of life



b) Prevent that Flood Causes Loss of life

Explore the triple neighbors of the anchor triple “Prevent that Flood Causes Loss of life” and apply the sentiment analysis.

TRIPLE NEIGHBORS	GOAL
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Loss of life
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Loss of life
Prevent that Flood Causes Property damage	Prevent that Flood Causes Loss of life
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Loss of life
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Loss of life
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Loss of life
Prevent that River overflow Causes Flood	Prevent that Flood Causes Loss of life
Sensors Detect Flood	Prevent that Flood Causes Loss of life
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Loss of life
Dam Avoids Flood	Prevent that Flood Causes Loss of life
Flood warning system Predicts Flood	Prevent that Flood Causes Loss of life
Cameras Detect Flood	Prevent that Flood Causes Loss of life
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Loss of life
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Loss of life
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Loss of life

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Loss of life	0.15536	0.70006	0.14458	NEUTRAL
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Loss of life	0.01621	0.98097	0.00282	NEUTRAL
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Loss of life	0.01536	0.51809	0.46655	NEUTRAL
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Loss of life	0.01216	0.48268	0.50516	CONTRADICTION
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Loss of life	0.01173	0.56807	0.4202	NEUTRAL
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Loss of life	0.01151	0.43703	0.55146	CONTRADICTION
Flood warning system Predicts Flood	Prevent that Flood Causes Loss of life	0.01093	0.9848	0.00427	NEUTRAL
Sensors Detect Flood	Prevent that Flood Causes Loss of life	0.00775	0.98452	0.00773	NEUTRAL
Cameras Detect Flood	Prevent that Flood Causes Loss of life	0.00747	0.98753	0.00501	NEUTRAL
Prevent that Flood Causes Property damage	Prevent that Flood Causes Loss of life	0.00673	0.26845	0.72482	CONTRADICTION
Dam Avoids Flood	Prevent that Flood Causes Loss of life	0.0059	0.99229	0.00181	NEUTRAL

Prevent that Siphoning Causes Flood	Prevent that Flood Causes Loss of life	0.00372	0.98981	0.00647	NEUTRAL
Prevent that River overflow Causes Flood	Prevent that Flood Causes Loss of life	0.00155	0.99648	0.00197	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Loss of life	0.00114	0.99649	0.00238	NEUTRAL
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Loss of life	0.00111	0.99718	0.00171	NEUTRAL

No entailment.
No triples related to loss of life due to flooding. It is difficult to make more abstract.
Stop the refinement.

Select the next anchor triple with highest entailment score:
Flood warning system Predicts Flood

c) Flood warning system Predicts Flood

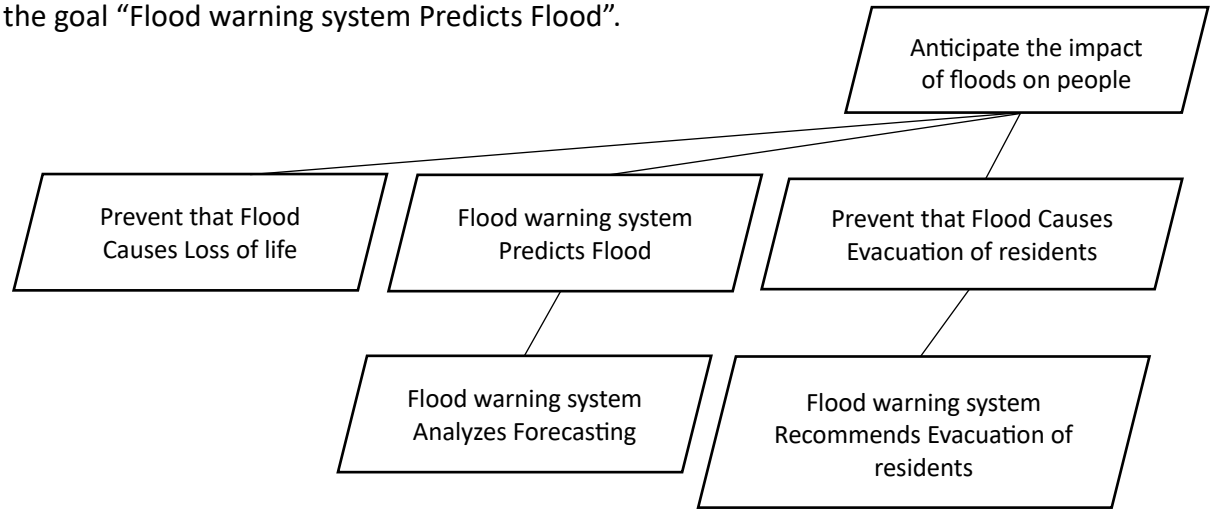
Explore the triple neighbors of "Flood warning system Predicts Flood" and apply the sentiment analysis.

TRIPLE NEIGHBORS	GOAL
Flood warning system Analyzes Forecasting	Flood warning system Predicts Flood
Flood warning system Generates Alerts	Flood warning system Predicts Flood
Flood warning system Recommends Evacuation of residents	Flood warning system Predicts Flood
Prevent that Flood Causes Environmental damage	Flood warning system Predicts Flood
Prevent that Flood Causes Loss of life	Flood warning system Predicts Flood
Prevent that Flood Causes Business disruption	Flood warning system Predicts Flood
Prevent that Flood Causes Property damage	Flood warning system Predicts Flood
Prevent that Flood Causes Traffic disruption	Flood warning system Predicts Flood
Prevent that Flood Causes Evacuation of residents	Flood warning system Predicts Flood
Prevent that Flood Causes Health and safety risks	Flood warning system Predicts Flood
Prevent that River overflow Causes Flood	Flood warning system Predicts Flood
Sensors Detect Flood	Flood warning system Predicts Flood
Prevent that Dam failure Causes Flood	Flood warning system Predicts Flood
Dam Avoids Flood	Flood warning system Predicts Flood
Cameras Detect Flood	Flood warning system Predicts Flood
Prevent that Siphoning Causes Flood	Flood warning system Predicts Flood
Prevent that Heavy rainfall Causes Flood	Flood warning system Predicts Flood
Prevent that Tsunami Causes Flood	Flood warning system Predicts Flood

Test the entailment.
Eliminate the triples that are already used and not relevant to flood prediction.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Flood warning system Recommends Evacuation of residents~~	~~Flood warning system Predicts Flood~~	~~**0.76855**~~	~~0.16382~~	~~0.06762~~	~~ENTAILMENT~~
Flood warning system Analyzes Forecasting	Flood warning system Predicts Flood	**0.58174**	0.34522	0.07305	ENTAILMENT
Flood warning system Generates Alerts	Flood warning system Predicts Flood	0.39238	0.54528	0.06234	NEUTRAL
Sensors Detect Flood	Flood warning system Predicts Flood	0.29798	0.58613	0.11589	NEUTRAL
Cameras Detect Flood	Flood warning system Predicts Flood	0.29682	0.52914	0.17404	NEUTRAL
Prevent that Tsunami Causes Flood	Flood warning system Predicts Flood	0.0624	0.8953	0.0423	NEUTRAL
Prevent that Dam failure Causes Flood	Flood warning system Predicts Flood	0.04879	0.90896	0.04226	NEUTRAL
Prevent that Siphoning Causes Flood	Flood warning system Predicts Flood	0.04744	0.89066	0.06191	NEUTRAL
Prevent that Flood Causes Traffic disruption	Flood warning system Predicts Flood	0.04296	0.92905	0.02799	NEUTRAL
Prevent that River overflow Causes Flood	Flood warning system Predicts Flood	0.03037	0.94772	0.02192	NEUTRAL
Prevent that Flood Causes Evacuation of residents	Flood warning system Predicts Flood	0.02889	0.95476	0.01634	NEUTRAL
Dam Avoids Flood	Flood warning system Predicts Flood	0.02796	0.43745	0.53459	CONTRADICTION
Prevent that Heavy rainfall Causes Flood	Flood warning system Predicts Flood	0.02538	0.961	0.01363	NEUTRAL
Prevent that Flood Causes Property damage	Flood warning system Predicts Flood	0.02113	0.96535	0.01352	NEUTRAL
Prevent that Flood Causes Business disruption	Flood warning system Predicts Flood	0.02112	0.96029	0.01859	NEUTRAL
Prevent that Flood Causes Environmental damage	Flood warning system Predicts Flood	0.01522	0.9715	0.01328	NEUTRAL
Prevent that Flood Causes Loss of life	Flood warning system Predicts Flood	0.01473	0.97541	0.00986	NEUTRAL
Prevent that Flood Causes Health and safety risks	Flood warning system Predicts Flood	0.00936	0.97448	0.01616	NEUTRAL

Refine the goal “Flood warning system Predicts Flood”.



Explore the triple neighbors of “Flood warning system Analyzes Forecasting” and apply the sentiment analysis. Remove the triples that are already used (in this branch).

TRIPLE NEIGHBORS	GOAL
~~Flood warning system Predicts Flood~~	~~Flood warning system Analyzes Forecasting~~
Flood warning system Generates Alerts	Flood warning system Analyzes Forecasting
Flood warning system Recommends Evacuation of residents	Flood warning system Analyzes Forecasting
Weather forecasting system Generates Forecasting	Flood warning system Analyzes Forecasting

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Flood warning system Predicts Flood~~	~~Flood warning system Analyzes Forecasting~~	~~0.42367~~	~~0.26419~~	~~0.31214~~	~~ENTAILMENT~~
Flood warning system Recommends Evacuation of residents	Flood warning system Analyzes Forecasting	0.04985	0.32959	0.62056	CONTRADICTION
Flood warning system Generates Alerts	Flood warning system Analyzes Forecasting	0.02521	0.30655	0.66824	CONTRADICTION
Weather forecasting system Generates Forecasting	Flood warning system Analyzes Forecasting	0.01009	0.02641	0.9635	CONTRADICTION

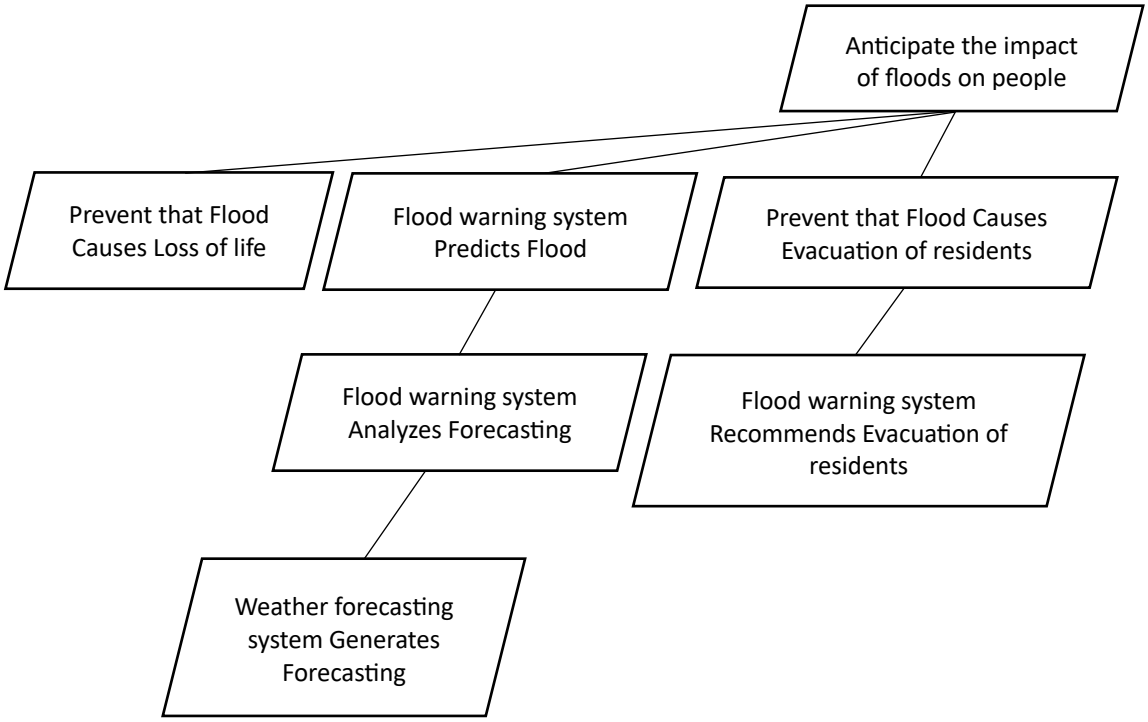
No entailment.

Relax the goal:
Flood warning system Analyzes Forecasting → **Obtain Forecasting**
(keep the object in the goal)

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Weather forecasting system Generates Forecasting	Obtain Forecasting	0.82942	0.11049	0.06009	ENTAILMENT
~~Flood warning system Predicts Flood~~	~~Obtain Forecasting~~	~~0.63306~~	~~0.25685~~	~~0.11009~~	~~ENTAILMENT~~
~~Flood warning system Recommends Evacuation of residents~~	~~Obtain Forecasting~~	~~0.50993~~	~~0.27654~~	~~0.21353~~	~~ENTAILMENT~~
~~Flood warning system Generates Alerts~~	~~Obtain Forecasting~~	~~0.48095~~	~~0.3347~~	~~0.18435~~	~~ENTAILMENT~~

Flood warning system Recommends Evacuation of residents → not focused on flood prediction.
Flood warning system Generates Alerts → not focused on flood prediction.



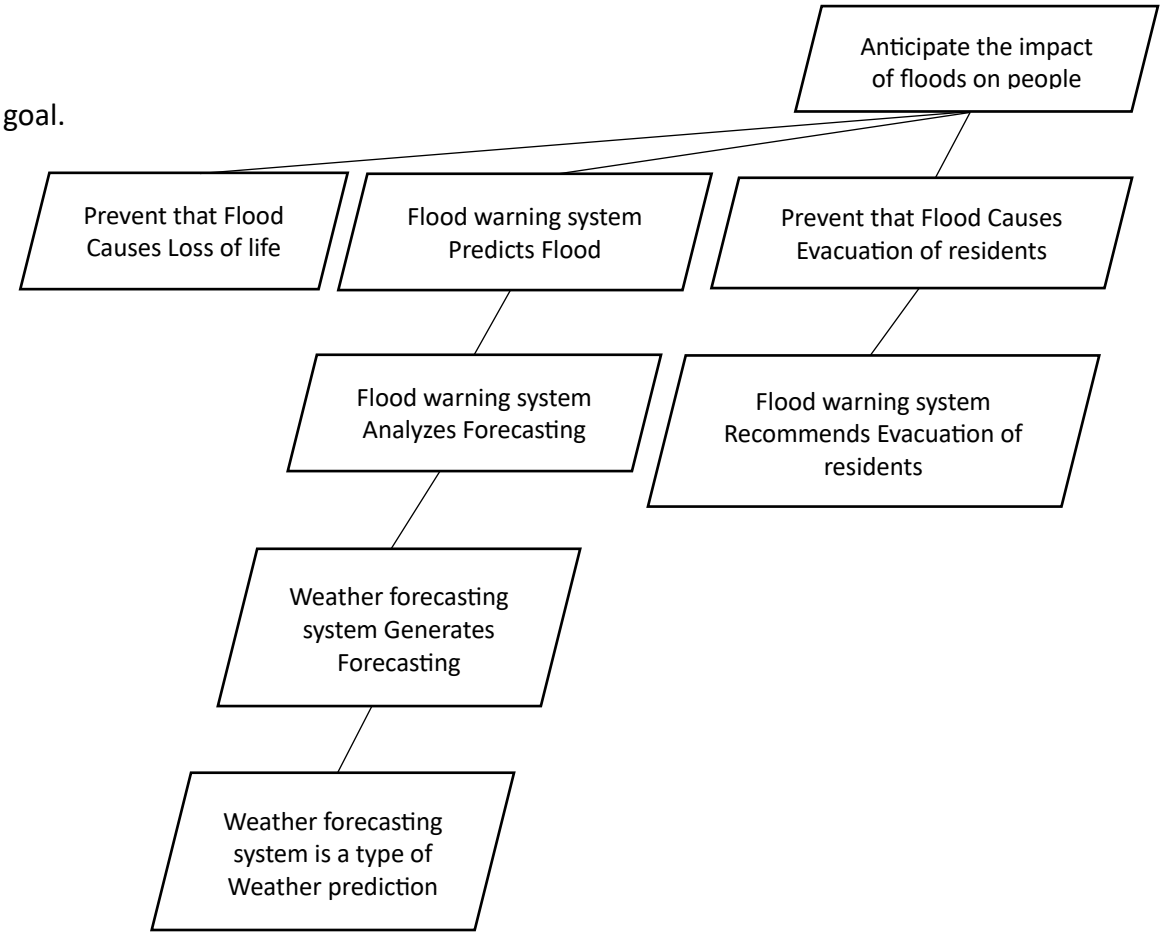
Explore the triple neighbors of "Weather forecasting system Generates Forecasting" (highest entailment score) and apply the sentiment analysis.
Remove the triples that are already used (in this branch).

TRIPLE NEIGHBORS	GOAL
Weather forecasting system Analyzes Data	Weather forecasting system Generates Forecasting
Weather forecasting system is a type of Weather prediction	Weather forecasting system Generates Forecasting
~~Flood warning system Analyzes Forecasting~~	~~Weather forecasting system Generates Forecasting~~

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Weather forecasting system is a type of Weather prediction	Weather forecasting system Generates Forecasting	0.83352	0.14557	0.02091	ENTAILMENT
Weather forecasting system Analyzes Data	Weather forecasting system Generates Forecasting	0.25562	0.29085	0.45353	CONTRADICTION
~~Flood warning system Analyzes Forecasting~~	~~Weather forecasting system Generates Forecasting~~	~~0.14854~~	~~0.08984~~	~~0.76162~~	~~CONTRADICTION~~

Refine the goal.



Explore the triple neighbors of "Weather forecasting system is a type of Weather prediction" and apply the sentiment analysis. Eliminate the triples that are already used (in this branch).

TRIPLE NEIGHBORS	GOAL
Weather forecasting system Analyzes Data	Weather forecasting system is a type of Weather prediction
~~Weather forecasting system Generates Forecasting~~	~~Weather forecasting system is a type of Weather prediction~~
Weather prediction Anticipates Heavy rainfall	Weather forecasting system is a type of Weather prediction

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Weather forecasting system Generates Forecasting~~	~~Weather forecasting system is a type of Weather prediction~~	~~0.83251~~	~~0.15345~~	~~0.01403~~	~~ENTAILMENT~~
Weather forecasting system Analyzes Data	Weather forecasting system is a type of Weather prediction	0.38863	0.53852	0.07285	NEUTRAL
Weather prediction Anticipates Heavy rainfall	Weather forecasting system is a type of Weather prediction	0.23617	0.72408	0.03975	NEUTRAL

No entailment.

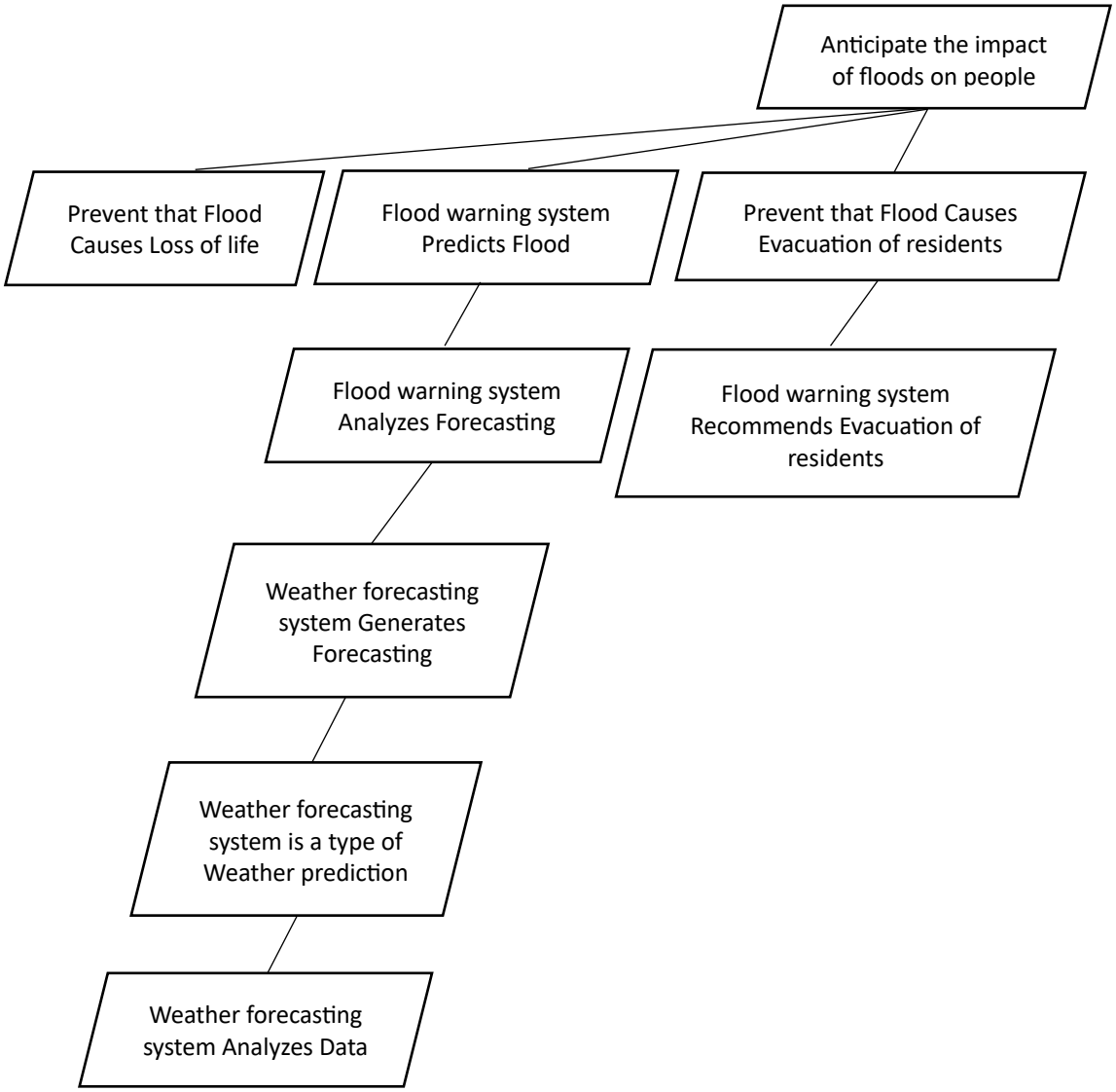
Relax the goal:
Weather forecasting system is a type of Weather prediction → **Weather prediction**

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Weather prediction Anticipates Heavy rainfall~~	~~Weather prediction~~	~~0.8954~~	~~0.05851~~	~~0.04609~~	~~ENTAILMENT~~
~~Weather forecasting system Generates Forecasting~~	~~Weather prediction~~	~~0.82887~~	~~0.13237~~	~~0.03876~~	~~ENTAILMENT~~
Weather forecasting system Analyzes Data	Weather prediction	0.76818	0.1611	0.07072	ENTAILMENT

Weather prediction Anticipates Heavy rainfall → too abstract.

Refine the goal “Weather forecasting system is a type of Weather prediction”.



Explore the triple neighbors of “Weather forecasting system Analyzes Data” and apply the sentiment analysis. Eliminate the triples that are already used (in this branch).

TRIPLE NEIGHBORS	GOAL
~~Weather forecasting system Generates Forecasting~~	~~Weather forecasting system Analyzes Data~~
~~Weather forecasting system is a type of Weather prediction~~	~~Weather forecasting system Analyzes Data~~
Sensors Generate Data	Weather forecasting system Analyzes Data
Cameras Generate Data	Weather forecasting system Analyzes Data

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Weather forecasting system is a type of Weather prediction~~	~~Weather forecasting system Analyzes Data~~	~~0.45716~~	~~0.38644~~	~~0.15641~~	~~ENTAILMENT~~
~~Weather forecasting system Generates Forecasting~~	~~Weather forecasting system Analyzes Data~~	~~0.26491~~	~~0.22943~~	~~0.50566~~	~~CONTRADICTION~~
Sensors Generate Data	Weather forecasting system Analyzes Data	0.00684	0.55787	0.4353	NEUTRAL
Cameras Generate Data	Weather forecasting system Analyzes Data	0.00524	0.467	0.52776	CONTRADICTION

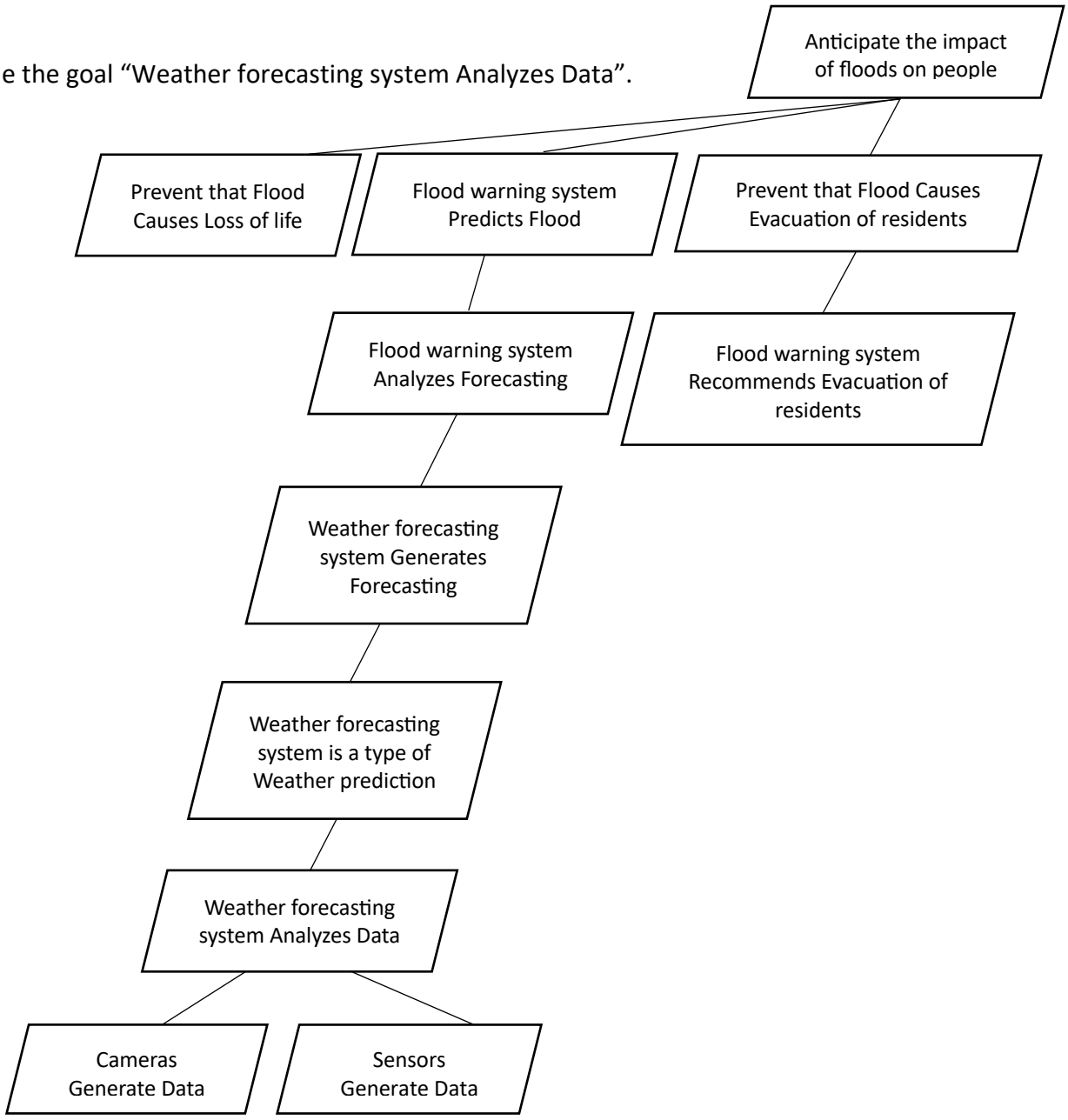
No entailment.

Relax the goal:
Weather forecasting system Analyzes Data → **Obtain Data**

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Cameras Generate Data	Obtain Data	0.89723	0.06839	0.03439	ENTAILMENT
Sensors Generate Data	Obtain Data	0.83576	0.08832	0.07591	ENTAILMENT
~~Weather forecasting system Generates Forecasting~~	~~Obtain Data~~	~~0.61847~~	~~0.28315~~	~~0.09837~~	~~ENTAILMENT~~
~~Weather forecasting system is a type of Weather prediction~~	~~Obtain Data~~	~~0.45241~~	~~0.4656~~	~~0.08199~~	~~NEUTRAL~~

Refine the goal “Weather forecasting system Analyzes Data”.



Explore the triple neighbors of “Cameras Generate Data” and apply the sentiment analysis. Remove the triples that are already used (in this branch).

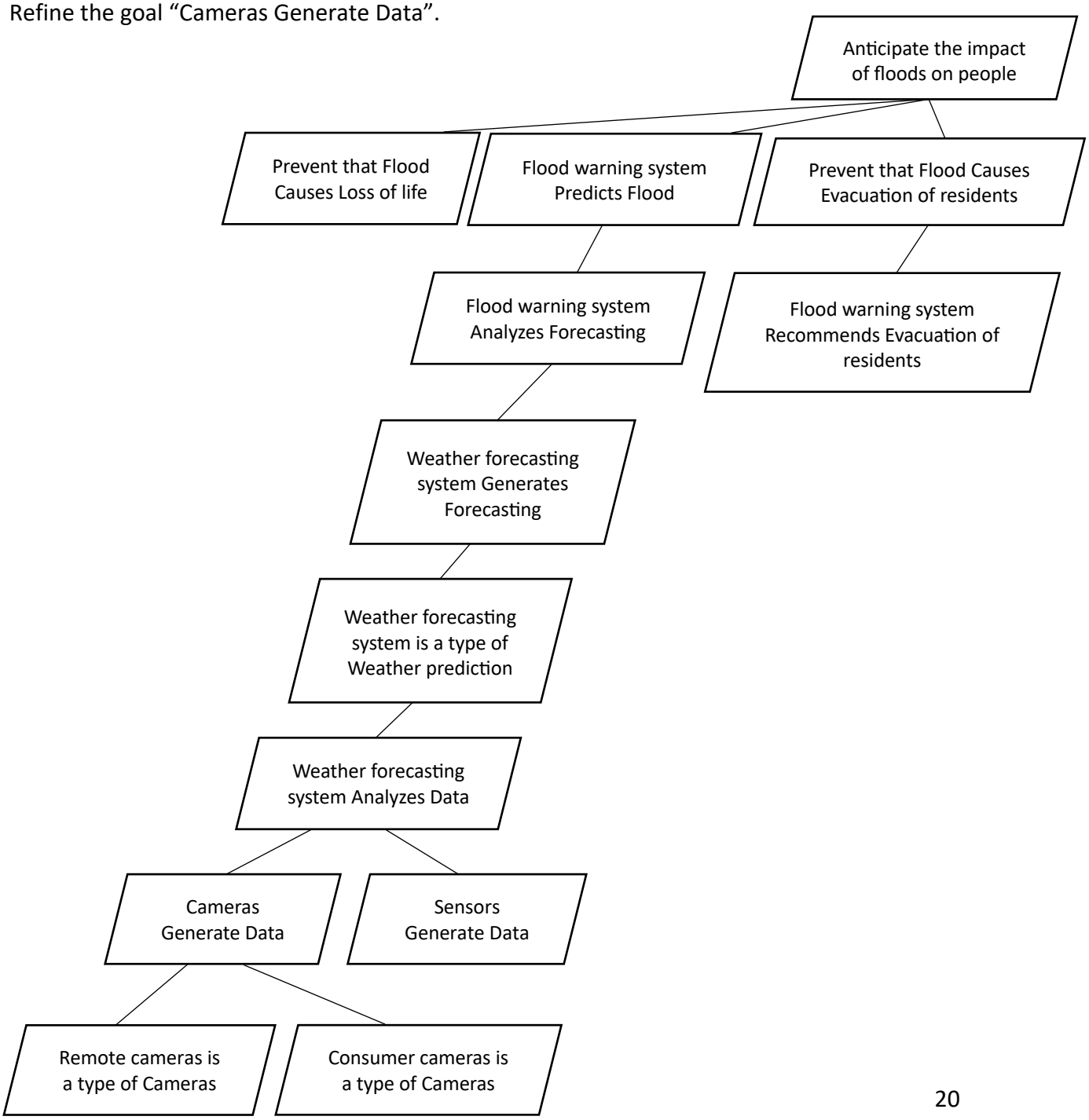
TRIPLE NEIGHBORS	GOAL
Cameras Detect Flood	Cameras Generate Data
Remote cameras is a type of Cameras	Cameras Generate Data
Consumer cameras is a type of Cameras	Cameras Generate Data
~~Sensors Generate Data~~	~~Cameras Generate Data~~
~~Weather forecasting system Analyzes Data~~	~~Cameras Generate Data~~

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
~~Cameras Detect Flood~~	~~Cameras Generate Data~~	~~0.74987~~	~~0.21113~~	~~0.039~~	~~ENTAILMENT~~
Remote cameras is a type of Cameras	Cameras Generate Data	0.5928	0.38731	0.01989	ENTAILMENT
Consumer cameras is a type of Cameras	Cameras Generate Data	0.51275	0.46888	0.01836	ENTAILMENT
~~Sensors Generate Data~~	~~Cameras Generate Data~~	~~0.0491~~	~~0.16023~~	~~0.79066~~	~~CONTRADICTION~~
~~Weather forecasting system Analyzes Data~~	~~Cameras Generate Data~~	~~0.04099~~	~~0.3018~~	~~0.65721~~	~~CONTRADICTION~~

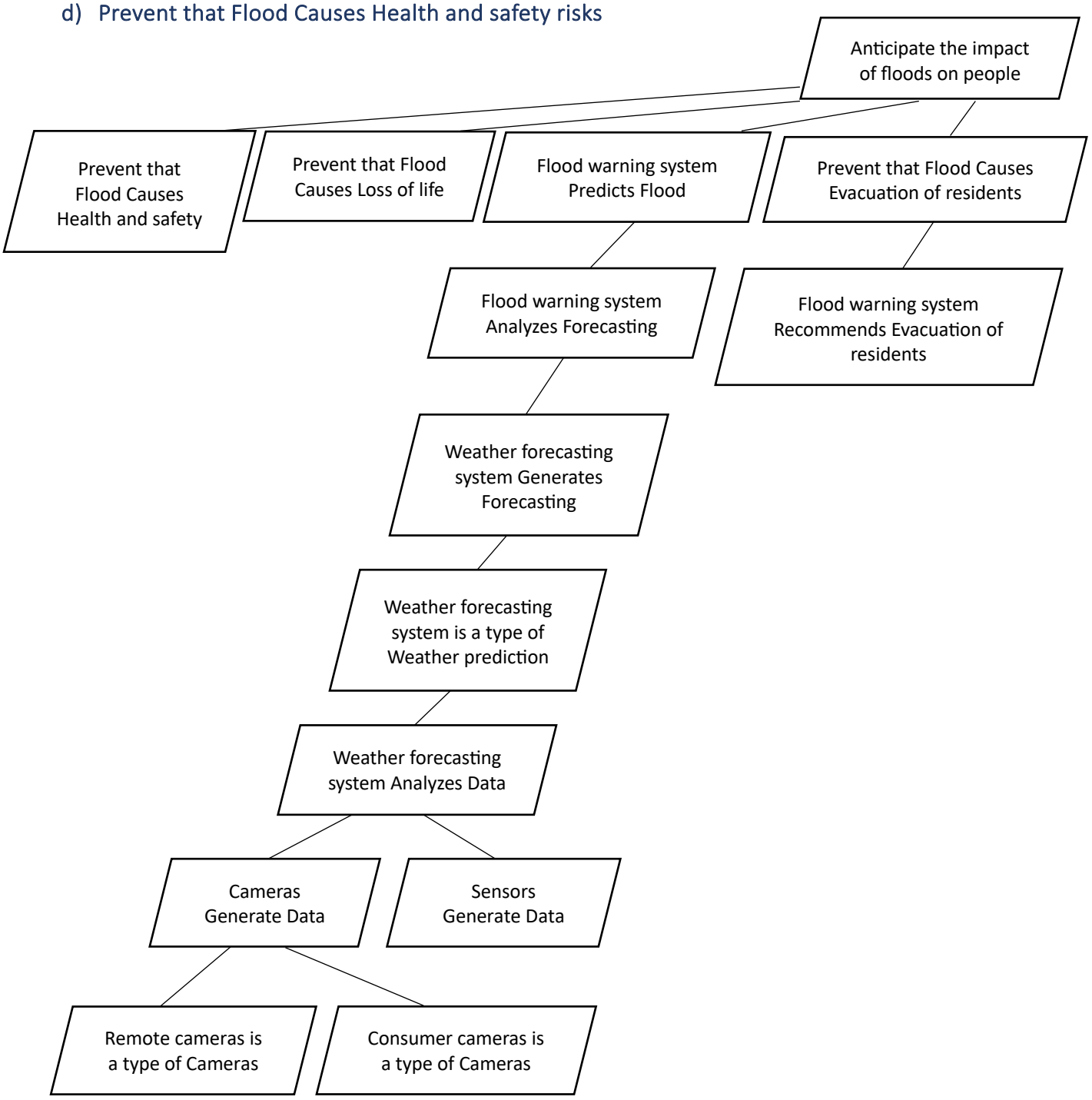
Cameras Detect Flood → too abstract

Refine the goal “Cameras Generate Data”.



Impossible to refine more.
All triple neighbors of “Remote cameras is a type of Cameras” and “Consumer cameras is a type of Cameras” are visited.
Stop the refinement.

d) Prevent that Flood Causes Health and safety risks



Explore the triple neighbors of “Prevent that Flood Causes Health and safety risks” and apply the sentiment analysis.

TRIPLE NEIGHBORS	GOAL
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Health and safety risks

Prevent that Flood Causes Loss of life	Prevent that Flood Causes Health and safety risks
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Health and safety risks
Prevent that Flood Causes Property damage	Prevent that Flood Causes Health and safety risks
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Health and safety risks
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Health and safety risks
Prevent that River overflow Causes Flood	Prevent that Flood Causes Health and safety risks
Sensors Detect Flood	Prevent that Flood Causes Health and safety risks
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Health and safety risks
Dam Avoids Flood	Prevent that Flood Causes Health and safety risks
Flood warning system Predicts Flood	Prevent that Flood Causes Health and safety risks
Cameras Detect Flood	Prevent that Flood Causes Health and safety risks
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Health and safety risks
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Health and safety risks
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Health and safety risks

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Health and safety risks	0.12413	0.69608	0.17979	NEUTRAL
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Health and safety risks	0.09282	0.7005	0.20668	NEUTRAL
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Health and safety risks	0.07441	0.48608	0.43952	NEUTRAL
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Health and safety risks	0.06036	0.80135	0.13829	NEUTRAL
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Health and safety risks	0.05312	0.91459	0.03229	NEUTRAL
Prevent that Flood Causes Property damage	Prevent that Flood Causes Health and safety risks	0.03894	0.4968	0.46426	NEUTRAL
Flood warning system Predicts Flood	Prevent that Flood Causes Health and safety risks	0.03754	0.95784	0.00461	NEUTRAL
Cameras Detect Flood	Prevent that Flood Causes Health and safety risks	0.03405	0.95569	0.01026	NEUTRAL
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Health and safety risks	0.02974	0.96619	0.00407	NEUTRAL
Sensors Detect Flood	Prevent that Flood Causes Health and safety risks	0.02111	0.96875	0.01014	NEUTRAL
Dam Avoids Flood	Prevent that Flood Causes Health and safety risks	0.01278	0.98578	0.00143	NEUTRAL
Prevent that River overflow Causes Flood	Prevent that Flood Causes Health and safety risks	0.01143	0.98607	0.0025	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Health and safety risks	0.01143	0.98627	0.0023	NEUTRAL

Prevent that Siphoning Causes Flood	Prevent that Flood Causes Health and safety risks	0.01085	0.98346	0.00569	NEUTRAL
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Health and safety risks	0.01012	0.98556	0.00432	NEUTRAL

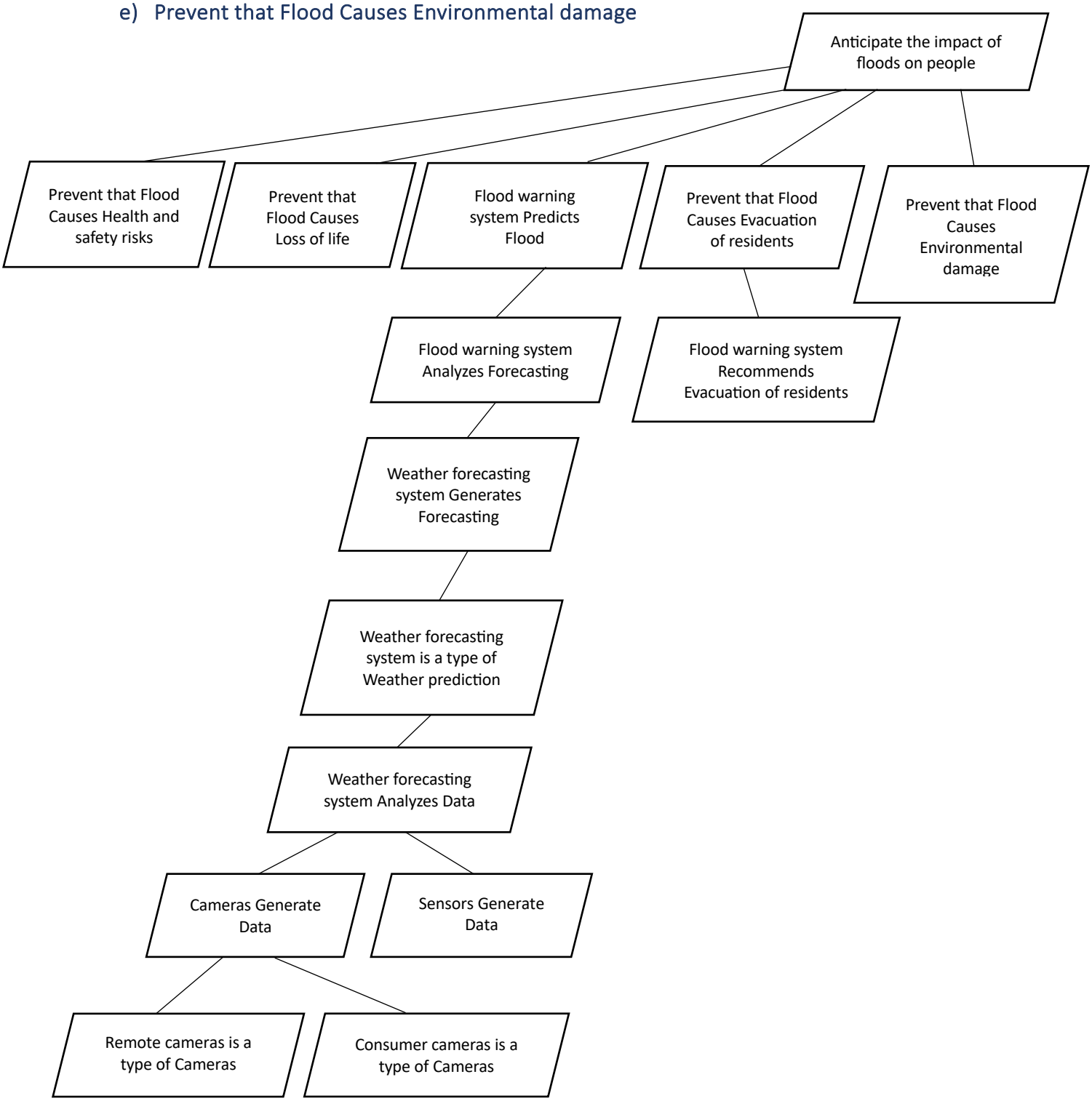
No entailment.

It is difficult to make the goal more abstract.

Stop the refinement.

Move to the next anchor triples.

e) Prevent that Flood Causes Environmental damage



Explore the triple neighbors and apply the sentiment analysis on triples.

TRIPLE NEIGHBORS	GOAL
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Environmental damage
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Environmental damage
Prevent that Flood Causes Property damage	Prevent that Flood Causes Environmental damage
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Environmental damage
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Environmental damage
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Environmental damage
Prevent that River overflow Causes Flood	Prevent that Flood Causes Environmental damage
Sensors Detect Flood	Prevent that Flood Causes Environmental damage
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Environmental damage
Dam Avoids Flood	Prevent that Flood Causes Environmental damage
Flood warning system Predicts Flood	Prevent that Flood Causes Environmental damage
Cameras Detect Flood	Prevent that Flood Causes Environmental damage
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Environmental damage
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Environmental damage
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Environmental damage

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Environmental damage	0.16927	0.26955	0.56118	CONTRADICTION
Flood warning system Predicts Flood	Prevent that Flood Causes Environmental damage	0.16315	0.8195	0.01735	NEUTRAL
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Environmental damage	0.15993	0.57692	0.26315	NEUTRAL
Dam Avoids Flood	Prevent that Flood Causes Environmental damage	0.13808	0.85462	0.0073	NEUTRAL
Prevent that River overflow Causes Flood	Prevent that Flood Causes Environmental damage	0.12732	0.85983	0.01285	NEUTRAL
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Environmental damage	0.12426	0.85439	0.02134	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Environmental damage	0.11374	0.87814	0.00812	NEUTRAL
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Environmental damage	0.10685	0.87607	0.01708	NEUTRAL
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Environmental damage	0.07953	0.17841	0.74206	CONTRADICTION
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Environmental damage	0.06315	0.90906	0.02779	NEUTRAL
Cameras Detect Flood	Prevent that Flood Causes Environmental damage	0.05162	0.92507	0.02332	NEUTRAL
Sensors Detect Flood	Prevent that Flood Causes Environmental damage	0.05129	0.92579	0.02292	NEUTRAL
Prevent that Flood Causes Property damage	Prevent that Flood Causes Environmental damage	0.03602	0.03493	0.92905	CONTRADICTION

Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Environmental damage	0.03206	0.08958	0.87836	CONTRADICTION
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Environmental damage	0.017	0.08956	0.89345	CONTRADICTION

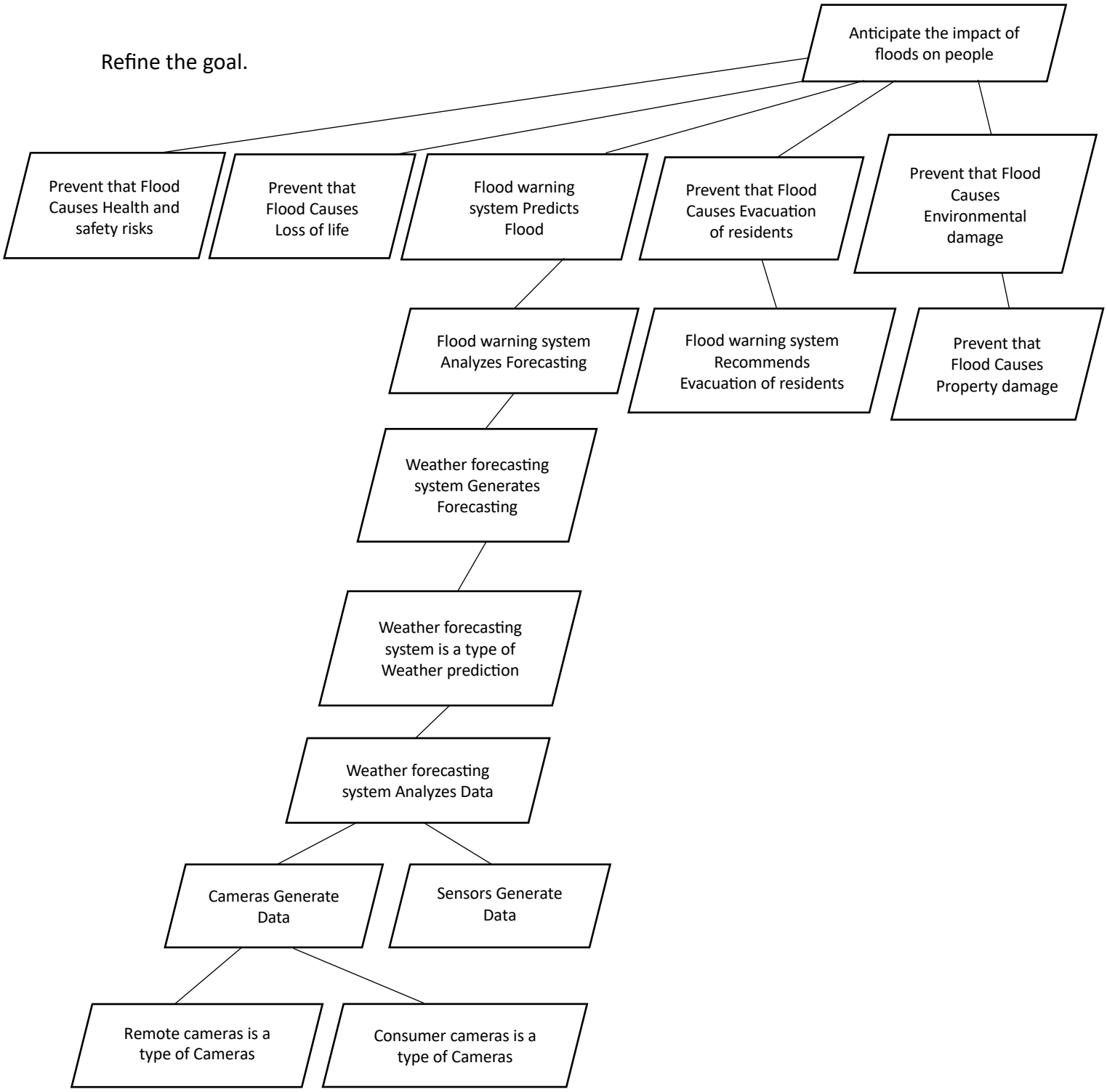
No entailment.

Relax the goal:

Prevent that Flood Causes Environmental damage → **Reduce environmental damage due to floods**

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Property damage	Reduce environmental damage due to floods	0.39956	0.23973	0.36071	ENTAILMENT
Prevent that Dam failure Causes Flood	Reduce environmental damage due to floods	0.39693	0.59267	0.0104	NEUTRAL
Prevent that Tsunami Causes Flood	Reduce environmental damage due to floods	0.39344	0.59089	0.01566	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Reduce environmental damage due to floods	0.39206	0.60085	0.00709	NEUTRAL
Prevent that River overflow Causes Flood	Reduce environmental damage due to floods	0.38735	0.60445	0.0082	NEUTRAL
Prevent that Flood Causes Health and safety risks	Reduce environmental damage due to floods	0.3634	0.5131	0.1235	NEUTRAL
Prevent that Siphoning Causes Flood	Reduce environmental damage due to floods	0.30862	0.6781	0.01328	NEUTRAL
Dam Avoids Flood	Reduce environmental damage due to floods	0.28795	0.70671	0.00533	NEUTRAL
Prevent that Flood Causes Loss of life	Reduce environmental damage due to floods	0.24603	0.5368	0.21717	NEUTRAL
Prevent that Flood Causes Traffic disruption	Reduce environmental damage due to floods	0.20737	0.55413	0.2385	NEUTRAL
Prevent that Flood Causes Evacuation of residents	Reduce environmental damage due to floods	0.19101	0.71267	0.09632	NEUTRAL
Prevent that Flood Causes Business disruption	Reduce environmental damage due to floods	0.15113	0.66596	0.18291	NEUTRAL
Flood warning system Predicts Flood	Reduce environmental damage due to floods	0.10709	0.85468	0.03823	NEUTRAL
Sensors Detect Flood	Reduce environmental damage due to floods	0.04074	0.92326	0.03601	NEUTRAL
Cameras Detect Flood	Reduce environmental damage due to floods	0.01894	0.9529	0.02816	NEUTRAL

Refine the goal.



Explore the triple neighbors of “Prevent that Flood Causes Property damage” and apply the sentiment analysis.

TRIPLE NEIGHBORS	GOAL
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Property damage
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Property damage
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Property damage
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Property damage
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Property damage

Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Property damage
Prevent that River overflow Causes Flood	Prevent that Flood Causes Property damage
Sensors Detect Flood	Prevent that Flood Causes Property damage
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Property damage
Dam Avoids Flood	Prevent that Flood Causes Property damage
Flood warning system Predicts Flood	Prevent that Flood Causes Property damage
Cameras Detect Flood	Prevent that Flood Causes Property damage
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Property damage
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Property damage
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Property damage
Bridge damage is a type of Property damage	Prevent that Flood Causes Property damage
Vehicles damage is a type of Property damage	Prevent that Flood Causes Property damage
Homes damage is a type of Property damage	Prevent that Flood Causes Property damage
Railroads damage is a type of Property damage	Prevent that Flood Causes Property damage
Buildings damage is a type of Property damage	Prevent that Flood Causes Property damage

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Prevent that Flood Causes Business disruption	Prevent that Flood Causes Property damage	0.33359	0.27574	0.39066	CONTRADICTION
Prevent that Flood Causes Evacuation of residents	Prevent that Flood Causes Property damage	0.11545	0.57208	0.31247	NEUTRAL
Flood warning system Predicts Flood	Prevent that Flood Causes Property damage	0.07852	0.91327	0.00821	NEUTRAL
Dam Avoids Flood	Prevent that Flood Causes Property damage	0.06797	0.92857	0.00346	NEUTRAL
Prevent that Tsunami Causes Flood	Prevent that Flood Causes Property damage	0.05941	0.92889	0.01169	NEUTRAL
Prevent that Siphoning Causes Flood	Prevent that Flood Causes Property damage	0.04281	0.93082	0.02636	NEUTRAL
Prevent that Flood Causes Loss of life	Prevent that Flood Causes Property damage	0.04124	0.12785	0.83092	CONTRADICTION
Prevent that Dam failure Causes Flood	Prevent that Flood Causes Property damage	0.04065	0.94668	0.01267	NEUTRAL
Prevent that Flood Causes Health and safety risks	Prevent that Flood Causes Property damage	0.03994	0.2344	0.72566	CONTRADICTION
Prevent that Flood Causes Environmental damage	Prevent that Flood Causes Property damage	0.03988	0.05615	0.90396	CONTRADICTION
Prevent that River overflow Causes Flood	Prevent that Flood Causes Property damage	0.03389	0.95463	0.01148	NEUTRAL
Sensors Detect Flood	Prevent that Flood Causes Property damage	0.03157	0.95694	0.01149	NEUTRAL
Cameras Detect Flood	Prevent that Flood Causes Property damage	0.03067	0.9594	0.00992	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Prevent that Flood Causes Property damage	0.02841	0.96775	0.00384	NEUTRAL
Prevent that Flood Causes Traffic disruption	Prevent that Flood Causes Property damage	0.02664	0.1596	0.81376	CONTRADICTION

Bridge damage is a type of Property damage	Prevent that Flood Causes Property damage	0.00985	0.90956	0.0806	NEUTRAL
Railroads damage is a type of Property damage	Prevent that Flood Causes Property damage	0.00719	0.38397	0.60884	CONTRADICTION
Vehicles damage is a type of Property damage	Prevent that Flood Causes Property damage	0.00661	0.72898	0.26441	NEUTRAL
Buildings damage is a type of Property damage	Prevent that Flood Causes Property damage	0.00391	0.92857	0.06752	NEUTRAL
Homes damage is a type of Property damage	Prevent that Flood Causes Property damage	0.00323	0.95502	0.04175	NEUTRAL

No entailment.

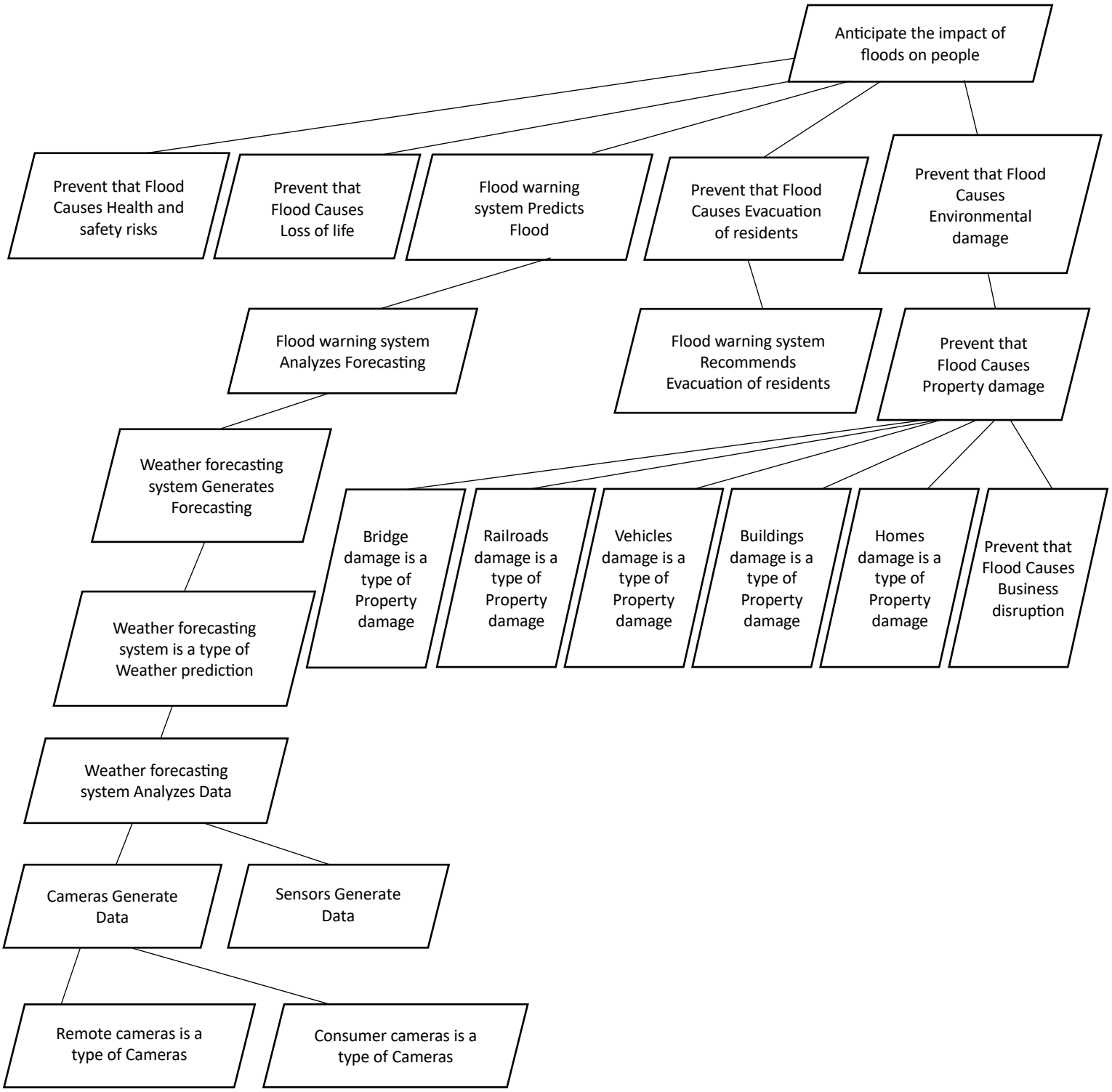
Relax the goal:

Prevent that Flood Causes Property damage → **Property damage**

Test the entailment.

PREMISE	HYPOTHESIS	E	N	C	NLI_LABEL
Bridge damage is a type of Property damage	Property damage	0.89772	0.07708	0.0252	ENTAILMENT
Railroads damage is a type of Property damage	Property damage	0.87966	0.08927	0.03107	ENTAILMENT
Vehicles damage is a type of Property damage	Property damage	0.86942	0.09429	0.0363	ENTAILMENT
Buildings damage is a type of Property damage	Property damage	0.8601	0.10026	0.03964	ENTAILMENT
Homes damage is a type of Property damage	Property damage	0.85746	0.09789	0.04466	ENTAILMENT
Prevent that Flood Causes Business disruption	Property damage	0.51538	0.35875	0.12586	ENTAILMENT
Prevent that Flood Causes Evacuation of residents	Property damage	0.31781	0.46307	0.21912	NEUTRAL
Cameras Detect Flood	Property damage	0.23153	0.56451	0.20396	NEUTRAL
Sensors Detect Flood	Property damage	0.21134	0.55492	0.23374	NEUTRAL
Prevent that Tsunami Causes Flood	Property damage	0.18326	0.48946	0.32728	NEUTRAL
Prevent that Siphoning Causes Flood	Property damage	0.18054	0.42149	0.39796	NEUTRAL
Prevent that Flood Causes Health and safety risks	Property damage	0.17339	0.35865	0.46796	CONTRADICTION
Prevent that Flood Causes Loss of life	Property damage	0.1698	0.25358	0.57662	CONTRADICTION
Prevent that Dam failure Causes Flood	Property damage	0.15461	0.47057	0.37482	NEUTRAL
Prevent that Flood Causes Traffic disruption	Property damage	0.14986	0.28108	0.56906	CONTRADICTION
Flood warning system Predicts Flood	Property damage	0.14956	0.80413	0.04632	NEUTRAL
Prevent that River overflow Causes Flood	Property damage	0.14658	0.67313	0.18029	NEUTRAL
Prevent that Heavy rainfall Causes Flood	Property damage	0.1446	0.78875	0.06665	NEUTRAL
Dam Avoids Flood	Property damage	0.10244	0.48777	0.40978	NEUTRAL
Prevent that Flood Causes Environmental damage	Property damage	0.0609	0.07317	0.86592	CONTRADICTION

Refine the goal “Prevent that Flood Causes Property damage”.



Impossible to refine more.
Prevent that Flood Causes Business disruption → No triples related to business disruption
No triples related to bridges, railroads, vehicles and homes damages.
Stop refinement.

→ Need to enrich the KG.